THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. How many terms of the AP $20, 16, 12, 8, \ldots$ give sum 0? 1

(a) 22 **(b)** 12 **(c)** 10 **(d)** 11

2. A square has diagonal $5\sqrt{2}$ cm. Its area (cm²) is: 1

(a) 10 **(b)** 50 **(c)** 100 **(d)** 25

3. The three points $(0,0)$, $(8,0)$ and $(3,7)$ enclose area: 1

(a) 42 **(b)** 28 **(c)** 56 **(d)** 14

4. If the 5th term of an AP is 11 and the 11th term is 5, then the 16th term is: 1

(a) 55 **(b)** 16 **(c)** 1 **(d)** 0

5. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 2 : 3$. The ratio of areas of $\triangle ADE$ to trapezium $DBCE$ is 4: 1

(a) 25 **(b)** 21 **(c)** 29 **(d)** 9

6. The point on the y-axis equidistant from $(1,2)$ and $(5,4)$ has $y =$ 1

(a) 9 **(b)** 3 **(c)** -9 **(d)** 18

7. The sum of the first 9 terms of an AP is 279, and its first term is 3. The 9th term is: 1

(a) 28 **(b)** 65 **(c)** 62 **(d)** 59

8. In a right triangle with hypotenuse $c = 13$ and one leg $a = 5$, the area (sq. units) is: 1

(a) 60 **(b)** 30 **(c)** $\frac{65}{2}$ **(d)** 78

9. The area of the triangle formed by $(11,0)$, $(0,4)$ and the origin, doubled, is: 1

(a) 22 **(b)** 88 **(c)** 44 **(d)** 15

10. In an AP, $S_{10} = 315$ and $S_9 = 261$. Then $a_{10} =$ 1

(a) 54 **(b)** 576 **(c)** 59 **(d)** 49

11. The altitude on the hypotenuse of the right triangle with legs 5 and 12 is of length (cm): 1

(a) 30 **(b)** $\frac{13}{2}$ **(c)** $\frac{17}{2}$ **(d)** $\frac{60}{13}$

12. If $P(x,y)$ is equidistant from $(2,4)$ and $(8,2)$ and lies on $y = 0$, then $x =$ 1

(a) 8 **(b)** 4 **(c)** 5 **(d)** -4

13. **Assertion (A):** The 10th term of the AP $2, 5, 8, \ldots$ is 29. 1
Reason (R): The nth term of an AP with first term a and common difference d is $a_n = a + nd$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** In a triangle ABC right-angled at B, with $AB = 6$ cm and $BC = 8$ cm, the hypotenuse $AC = 10$ cm. 1
Reason (R): The Pythagoras theorem holds only for isosceles right-angled triangles.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The mid-point of the segment joining $(2, -3)$ and $(8, 5)$ is $(5, 1)$. 1
Reason (R): The mid-point of the segment joining (x_1, y_1) and (x_2, y_2) is $\left(\frac{x_1 - x_2}{2}, \frac{y_1 - y_2}{2}\right)$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the value of k for which the points $(1, 5)$, $(2, 8)$ and $(4, k)$ are collinear. 2

17. In an AP, $a_3 + a_7 = 34$ and $a_2 + a_9 = 37$. Find a and d. 2

18. Find the altitude on the hypotenuse of the right triangle with legs 3 cm and 4 cm. 2

19. The vertices of a triangle are $(0, 0)$, $(6, 0)$, $(0, 8)$. Find the length of the median from the first vertex. 2

20. The 5th term of an AP is 12 and the 12th term is 5. Find the 17th term. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. $\triangle ABC$ is right-angled at B with $AB = 3$ cm and $BC = 4$ cm; $BD \perp AC$. Find AD, DC and BD. 3

22. Find the values of y for which the distance between $P(8, y)$ and $Q(2, 3)$ is 10 units. 3

23. In an AP, $S_5 = 50$ and $S_9 = 162$. Find the AP. 3

24. In trapezium $ABCD$, $AB \parallel DC$ and the diagonals meet at O. If $AB = 6$ cm and $DC = 10$ cm, find $AO : OC$ and the ratio $ar(AOB) : ar(COD)$. 3

OR

Show that the points $A(0, 0)$, $B(6, 8)$, $C(16, 8)$, $D(10, 0)$ are the vertices of a rhombus. 3

25. Find the sum of the first 20 terms of the AP whose 2nd term is 14 and 7th term is 39. 3

OR

Find the altitude of an equilateral triangle of side 8 cm, in terms of $\sqrt{3}$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** At the same time of the day, a girl of height 1.5 m casts a shadow 2 m long, while a tower casts a shadow 24 m long on level ground. 5

(i) Name the similarity criterion that makes the two triangles formed (object, shadow, sun's ray) similar.

(ii) Find the height of the tower.

(iii) At that time, a pole 6 m high will cast a shadow of what length?

(iv) Find the distance from the top of the tower to the tip of its shadow.

27. A ladder has 31 rungs whose lengths decrease uniformly from 90 cm at the bottom to 30 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. 5

OR

Two ships leave a port at the same time; one sails 50 km due north and the other 120 km due west. Find the distance between them and the ratio of this distance to the one if both had sailed only half as far. 5

END OF QUESTION PAPER Total: 50 Marks

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